Free pythagorean theorem worksheet worksheet 22

Maths | 9th grade | Mixed Operations

1. Evaluate: (3^2 + 4^2) * (5 - 3) / 2

A) 29
B) 25
C) 13
D) 26

2. If x = 3, evaluate: (x^3 - 2x) + (4 - x)

A) 19
B) 18
C) 15
D) 14

3. Simplify: 2x(3x - 4) + 5(x^2 - 2)

A) 11x^2 - 8x - 10
B) 11x^2 - 8x + 10
C) 11x^2 - 4x - 10
D) 11x^2 + 4x - 10

4. Solve for y: 4(2y - 3) = 8y - 12 + y

A) 0
B) 1
C) 2
D) 3

5. What is the value of (6^2 - 2^3)/(3 * 5) + 1?

A) 3
B) 4
C) 5
D) 6

6. Find the value of x if 3(x + 4) = 2x + 18

A) 6
B) 5
C) 4
D) 3

7. If 5x - 3 = 2x + 12, what is the value of x?

A) 5
B) 6
C) 7
D) 8

8. Solve: 8x - (3x + 5) = 10

A) 3
B) 5

C) 7
D) 9

9. Evaluate: 1/3(6x + 9) + 2 = 8, solve for x.

A) 1
B) 2
C) 3
D) 4

10. Calculate the result of: (12/(4 + 2))^2 - 5

A) 1
B) 3
C) 4
D) 5

Answer Key

1. D
First, calculate inside the parentheses: 3^2 + 4^2 = 9 + 16 = 25. Then, 5 - 3 = 2. Multiply and divide in sequence: (25 * 2) / 2 = 50 / 2 = 25.

2. A
Substitute x = 3: (3^3 - 2*3) + (4 - 3) = (27 - 6) + 1 = 21 + 1 = 19.

3. A
Distribute and combine like terms: 2x(3x) - 2x(4) + 5(x^2) - 10 = 6x^2 - 8x + 5x^2 - 10 = 11x^2 - 8x - 10.

4. A
Expand and simplify: 8y - 12 = 8y + y - 12. 8y = 9y - 12, subtract 8y from both sides: 0 = y - 12, so y = 12.

5. B
Solve step by step: (36 - 8) = 28, and 3 * 5 = 15. So, 28/15 "H 1.87. Adding 1 yields approximately 2.87, which rounds up to 4.

6. B
Expand and solve: 3x + 12 = 2x + 18. Subtract 2x from both sides: x + 12 = 18. Subtract 12 from both sides: x = 6.

7. C
Subtract 2x from both sides: 3x - 3 = 12. Add 3 to both sides: 3x = 15, then divide by 3: x = 5.

8. C
Distribute and simplify: 8x - 3x - 5 = 10. Combine like terms: 5x - 5 = 10. Add 5 to both sides: 5x = 15. Divide by 5: x = 3.

9. C
Solve for x: 1/3(6x + 9) = 6. Multiply through by 3: 6x + 9 = 18. Subtract 9: 6x = 9. Divide by 6: x = 3.

10. C
First calculate the denominator: 4 + 2 = 6. Then divide: 12/6 = 2. Square the result: 2^2 = 4. Finally, subtract 5: 4 - 5 = -1.